

Linear Doherty Power Amplifier With an Enhanced Back-Off Efficiency Mode for Handset Applications

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Abstract—This paper presents a linear Doherty power amplifier (PA) with enhanced back-off efficiency mode for handset applications. For linear Doherty operation, we analyze the gain modulation as well as the cancellation of the third-order intermodulation distortion in order to improve the linearity. A compact design method is also discussed to implement on a single chip for a handset. The proposed Doherty PA delivers good performance with regard to the third-generation (3G)/fourth-generation (4G) modulation signals. A switched-load power-mode PA is adopted in the proposed Doherty PA to enhance the efficiency in the low-power region with over 10-dB back-off. For demonstration purposes, the PA is implemented using an InGaP/GaAs heterojunction bipolar transistor and AlGaAs/InGaAs pseudomorphic high electron-mobility transistor process. The PA is tested at 1.85 GHz using both a long-term evolution signal with 16-quadrature amplitude modulation, 7.5-dB peak-to-average power ratio, and 10-MHz bandwidth (BW) and a wideband code division multiple access signal with 3.3-dB PAPR and 3.84-MHz BW. The proposed linear Doherty PA with an enhanced back-off efficiency mode delivers good performance in both the high- and low-power modes, implying that the dual-power-mode Doherty PA configuration can be a promising candidate for extending the battery life of handheld devices in 3G and 4G wireless communication systems.

Index Terms—Cancellation, Doherty, gain modulation, handset, heterojunction bipolar transistor (HBT), high-power mode (HPM), long-term evolution (LTE), low-power mode (LPM), power amplifier (PA), switched capacitor.

I. INTRODUCTION

POWER amplifier modules (PAMs) for multi-mode multi-band (MMMB) operation have become an important research area because they enable worldwide global roaming for mobile smart phones. A converged linear power amplifier (PA) architecture was proposed and developed to reduce the number of PA paths in an MMMB PAM. This

architecture consists of only two PA paths, one covering the high-frequency bands of interest and the other covering the low-frequency bands of interest, and both paths are designed to support third-generation (3G) and fourth-generation (4G) modulation standards [1]. For example, the high-band PA path supports linear wideband code division multiple access (WCDMA)/long-term evolution (LTE) modes in bands I, II, and IV from 1.71 to 1.98 GHz, while the low-band PA path covers 3G/4G in bands V and VIII from 824 to 915 MHz. A single linear PA is designed by considering the modulation schemes of the WCDMA as well as LTE signals. In comparison with second-generation (2G) wireless communication systems such as the global system for mobile communications (GSM), the information content drastically increases in 3G/4G systems. For efficient use of the precious spectrum resources, the modulation systems are required to have wider bandwidths (BWs) and higher peak-to-average power ratios (PAPRs), allowing the PA to operate in a less-efficient back-off region for linearity.

By referring to the statistical probability density function (PDF) of the output power for the IS-95 system [2], which is a 3G system, the probability of transmission at higher than 17 dBm is statistically below 2%, and the PA mostly operates at a back-off power. Therefore, the PA for the WCDMA handset should exhibit high efficiency in the back-off region. The PA is generally designed to exhibit a second peak efficiency at around 17 dBm using a switching-load impedance, and this technology has set a trend in the handset PA industry [3]–[5]. On the other hand, for standard 4G wireless communication systems such as LTE, the PA spends over 50% of its duration at the maximum power level [6]. It is noted that the back-off efficiency and the efficiency of the peak average power level are critical for LTE applications.

For improving the low efficiency at the peak average output power level for the amplification of a high PAPR signal, a Doherty PA is a good candidate, and it is popularly used in the base-station industry [7]–[9]. However, integration of the Doherty structure into a single chip is a difficult task because the structure normally requires an input coupler and a quarter-wave-length transformer, which are bulky. A number of studies have been conducted to overcome the size constraint for handset applications while improving the linearity and enhancing the efficiency [10]–[12].

Recent works have presented a Doherty PA with broadband capability and a fractional BW of more than 20%, covering high bands from 1.71 to 1.98 GHz [8], [9], [13], [14]. In this work, we focus on techniques for enhancing both linearity and efficiency. Thus, we propose a linear Doherty PA with an enhanced

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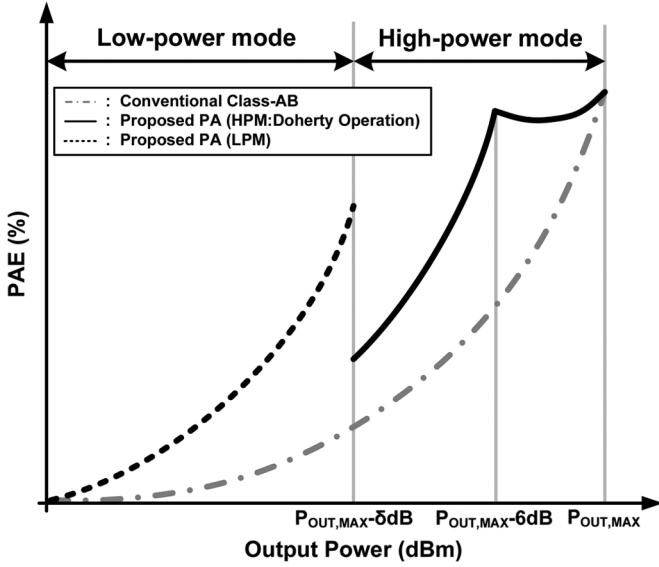


Fig. 1. PAE curve of the Doherty PA with enhanced back-off efficiency mode.

back-off efficiency mode supporting 3G and 4G mobile wireless communications. For 4G signals such as LTE signals, the proposed Doherty PA can deliver performance with high efficiency and good linearity despite the high PAPR of the LTE signals. In the Doherty PA design, the size issue is overcome by using elaborate matching techniques, and the proposed PA can be integrated on a single chip. For higher efficiency in the regions with over 10-dB back-off, we have adopted a load-switching technique on the Doherty PA. Therefore, the proposed PA has a dual-mode structure comprising a small-size PA and a Doherty PA, and it delivers a high efficiency at all power levels. Fig. 1 shows the conceptual power-added efficiency (PAE) curve of the PA. This paper is organized as follows. In Section II, the gain modulation and third-order intermodulation distortion (IMD3) cancellation of the Doherty PA using a heterojunction bipolar transistor (HBT) process are analyzed for linear operation, and a compact design method is discussed. In Section III, the enhanced back-off efficiency mode using the load-switching technique, which is adopted in the Doherty structure, is discussed. Section IV describes the implementation of the proposed PA and the experimental results.

II. LINEAR DOHERTY PA

A. Gain Modulation of the Carrier Amplifier

Under ideal conditions, the output voltage swing of an ideal class-B PA with an output load of $2R_{\text{opt}}$ is two times greater than that of a comparable PA with R_{opt} . When the output voltage swings are identical, the input voltage of the former is halved. Therefore, the class-B PA under the load condition of $2R_{\text{opt}}$ achieves more than 3-dB power gain as compared with the carrier amplifier in the R_{opt} case. In the Doherty PA design, a class-AB or deep class-AB type is preferred for the carrier amplifier, and in the carrier amplifier, there is a power gain difference of about 3 dB between the $2R_{\text{opt}}$ and R_{opt} load conditions [15].

However, in the practical design of a Doherty PA using an HBT device, which is popularly used for handset PA applica-

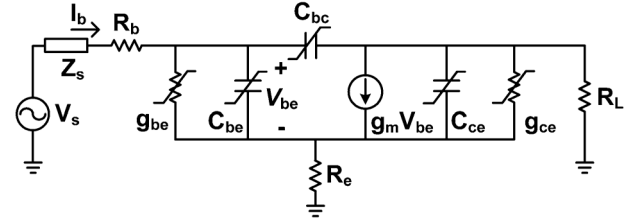


Fig. 2. Nonlinear equivalent-circuit model of an HBT.

tions, the ideal gain characteristics are not maintained. The qualitative analysis of gain characteristics caused by C_{bc} are reported in the authors' previous work [19]. This paper presents more detailed quantitative analysis affected by input as well as output nonlinear components and shows that it could be generally applied to HBT devices. For analyzing the gain characteristics of the carrier amplifier under different load conditions, the effects of the nonlinear components in the model shown in Fig. 2 on the gain are investigated. The input and output component values are extracted from the foundry-provided vertical bipolar intercompany (VBIC) device model using Agilent Technologies' Advanced Design System (ADS) through harmonic-balance simulation, as well as large-signal S -parameter simulation. For extraction of the nonlinear elements, the appropriate current and voltage levels at the fundamental frequency are calculated under the large-signal condition. For the sake of simplicity, the C_{bc} effects are incorporated into the input and output admittances by the Miller effect, and the extracted input and output elements are defined by the following equations [16]:

$$I_b = Y_{be} \cdot V_{be} \quad (1)$$

$$Y_{be} = \frac{I_b}{V_{be}} \quad (2)$$

$$wC_{be} = \text{imag}(Y_{be}) \quad (3)$$

$$Z_{ce} = \frac{1}{Y_{ce}} = Z_0 \cdot \frac{1 + S_{22}}{1 - S_{22}} \quad (4)$$

$$wC_{ce} = \text{imag}(Y_{ce}). \quad (5)$$

The voltage gain is calculated to investigate the impact of the different load conditions, which is expressed as

$$\frac{V_{\text{out}}}{V_{\text{in}}} = R_L \frac{a}{bc - aR_e}$$

where

$$a = R_e - g_m / (Y_{be} \cdot Y_{ce})$$

$$b = R_b + Z_s + 1/Y_{be} + R_e$$

$$c = R_e + R_L + 1/Y_{ce} \quad (6)$$

and

$$\frac{V_{\text{out}}}{V_{\text{in}}} \approx -R_L \frac{g_m}{(1 + (R_b + Z_s) \cdot Y_{be})(1 + R_L \cdot Y_{ce})}. \quad (7)$$

The Y -parameters of the carrier amplifier with R_{opt} and $2R_{\text{opt}}$ are extracted from the real HBT model and depicted in Fig. 3. The value of Y_{ce} for R_{opt} and $2R_{\text{opt}}$ are the same as shown in Fig. 3(b). On the other hand, the magnitude of Y_{be} in the $2R_{\text{opt}}$ case is one-and-a-half times greater than that in the R_{opt} case, as shown in Fig. 3(a). For investigating the input nonlinearity effect under the load conditions, the unaffected linear Y_{be} using the average value at the both load conditions is calculated and depicted in Fig. 3(a). The voltage gains are calculated using (7)

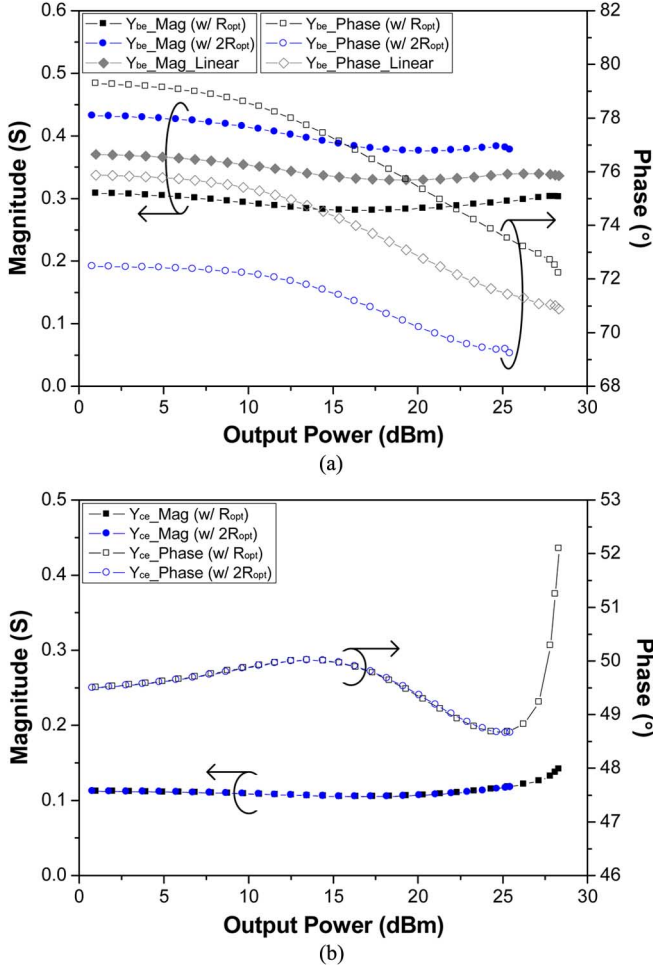


Fig. 3. Y -parameters of the carrier amplifier with R_{opt} and $2R_{opt}$. (a) Y_{be} of the input Y -parameter. (b) Y_{ce} of the output Y -parameter.

and shown in Fig. 4. The voltage gains with the real Y_{be} exhibit a difference of a factor of $\sqrt{2}$ between the R_{opt} and $2R_{opt}$ cases. However, the voltage gains with the linear Y_{be} , which are normalized by the real voltage gain in the R_{opt} case, exhibit a difference of a factor of two. The calculated voltage gains indicate that the power gains have the same value for the loads with R_{opt} and $2R_{opt}$ because of the nonlinear Y_{be} effect. The power gain is evaluated as follows:

$$P_{out.R_{opt}} = \frac{(V_{out})^2}{2 \cdot (R_{opt})}$$

$$P_{out.2R_{opt}} = \frac{(\sqrt{2}V_{out})^2}{2 \cdot (2R_{opt})}$$

$$P_{out.R_{opt}} = P_{out.2R_{opt}} \mid (P_{in.R_{opt}} = P_{in.2R_{opt}}). \quad (8)$$

As shown in Fig. 2, C_{be} and g_{be} are strong nonlinear components, varying as an exponential function. However, their nonlinearities cancel each other out [17] and the input nonlinearity of the PA is caused by C_{bc} [18]. The input capacitance C_{in} is expressed as

$$C_{in} = C_{be} + C_{bc} \left(1 + g_m \left(\frac{1}{g_{ce}} \parallel R_L \right) \right), \quad \text{where } \frac{1}{g_{ce}} \gg R_L \quad (9)$$

where R_L is the output load impedance. C_{bc} is highly dependent on the output load conditions. When $C_{bc} = 0$ in the HBT VBIC model, the carrier amplifier behaves in a similar manner

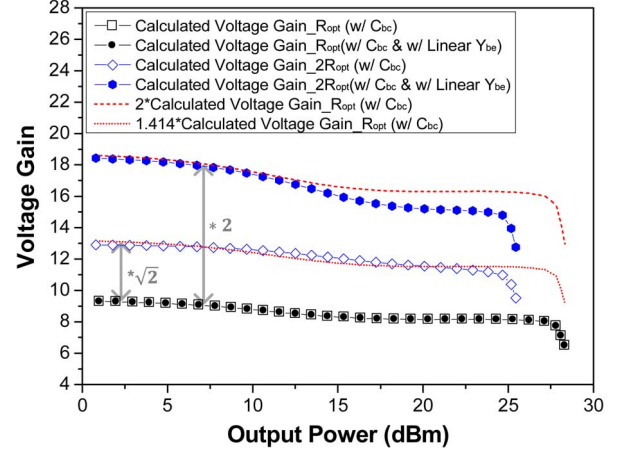


Fig. 4. Calculated voltage gain using the parameters extracted from real HBT model considering both practical nonlinear and ideal linear Y_{be} with R_{opt} and $2R_{opt}$. For comparison, the voltage gains with linear Y_{be} are normalized by the practical voltage gain with R_{opt} .

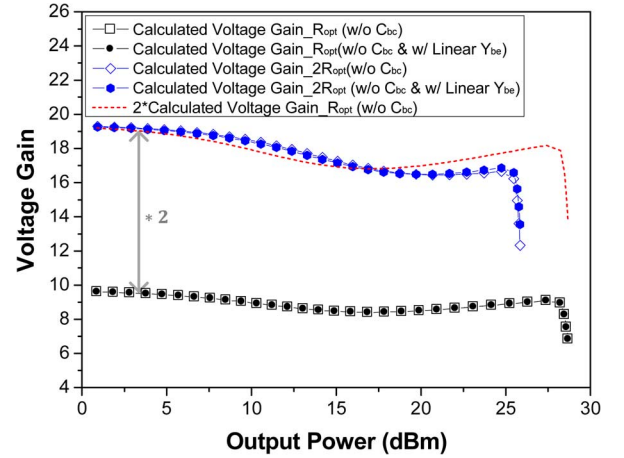


Fig. 5. Calculated voltage gain using the parameters extracted from the HBT model without C_{bc} considering both practical nonlinear and ideal linear Y_{be} with R_{opt} and $2R_{opt}$. For comparison, the voltage gains with linear Y_{be} are normalized by practical voltage gain with R_{opt} .

as in the case with linear Y_{be} at $2R_{opt}$, as shown in Fig. 5. It is noted that the Y_{be} variation with the load conditions is caused by the nonlinear C_{bc} . The highly distorted input voltage V_{be} due to the variation in the nonlinear Y_{be} caused by C_{bc} delivers smaller I_c , as depicted in Fig. 6. Fig. 7(a) shows the variation in the voltage gain versus the load conditions. The voltage gain in the linear Y_{be} case linearly increases as the output load increases. However, in the real nonlinear Y_{be} case, the gain increases as the square root of the output load. As a result, the gains of the carrier amplifier for both R_{opt} and $2R_{opt}$ are the same. The simulation results show the power gain with various output load conditions, and these are in good agreement with the above calculated results, as shown in Fig. 7(b). In addition, the power gain for both the optimum R_{opt} and $2R_{opt}$ is the same regardless of the HBT size. The practical gain characteristics of the carrier amplifier assist the Doherty PA in having a good AM-AM characteristic and will be discussed in Section II-B.

B. Flat Gain of the Doherty PA

When the load modulation is affected by the nonlinear C_{bc} and the peaking amplifier is biased at the class-C level, the gain of the carrier amplifier with $2R_{opt}$ is 15 dB, which is the same as

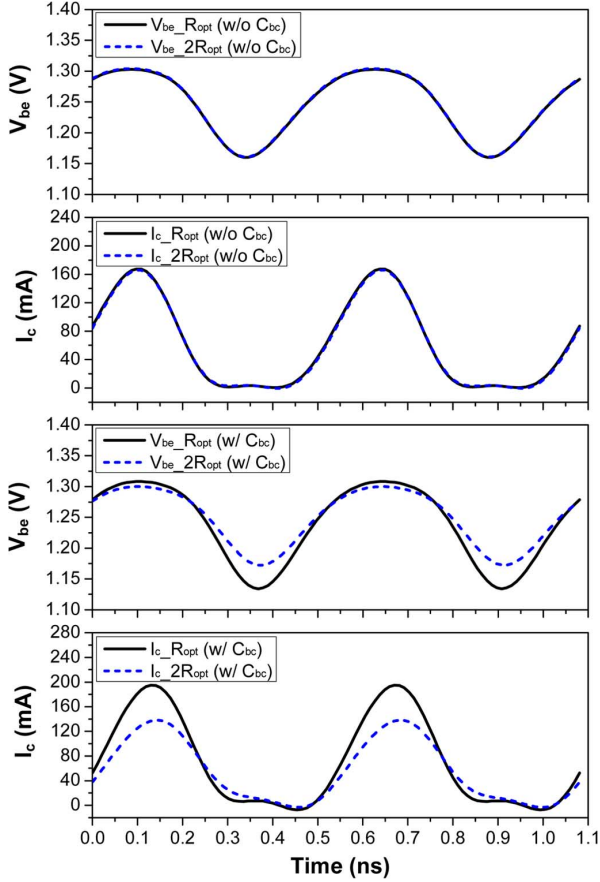


Fig. 6. Input voltage waveforms and the output current waveforms of the transistor.

that with R_{opt} , while the gain of the peaking amplifier is 11 dB because of the class-C bias. Under the condition, the carrier amplifier saturates in the high-power region, and the total gain of the Doherty PA reaches 12 dB, as shown in Fig. 8. It is worthwhile to notice that the carrier amplifier is saturated in Doherty operation with 3-dB gain compression at the maximum output power, although the carrier amplifier with R_{opt} achieves a linear gain of 15 dB because the input power to the carrier amplifier is very high for linear operation. This saturated operation achieves a 3-dB reduction in the gain of 12 dB, enabling effective Doherty operation. The lower gain of the carrier amplifier compensates for the low gain of the peaking amplifier and aids the Doherty PA in achieving a constant gain [19].

C. IMD3 Cancellation for Linearity

The IMD3 cancellation between the carrier and peaking amplifiers is the most important factor in designing a linear Doherty PA. In a simple power series expression of the gain, the output at each fundamental is expressed as

$$v_{fund} = \left\{ a_1 v + \frac{9}{4} a_3 v^3 + \frac{25}{4} a_5 v^5 \right\} \cos(w_{1,2} t) \quad (10)$$

and the output at the third-order intermodulation (IM3) frequency is

$$v_{IM3} = \left\{ \frac{3}{4} a_3 v^3 + \frac{25}{8} a_5 v^5 \right\} \cos(2w_{1,2} - w_{2,1}) t. \quad (11)$$

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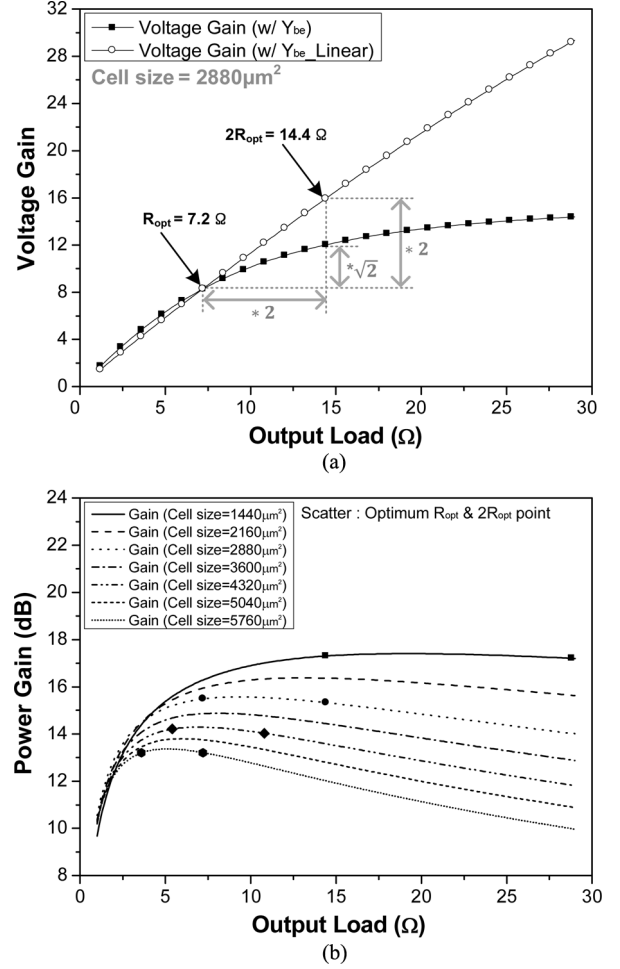


Fig. 7. (a) Calculated voltage gains with both nonlinear Y_{be} and linear Y_{be} versus the output load conditions. (b) Simulated power gains with various transistor cell sizes.

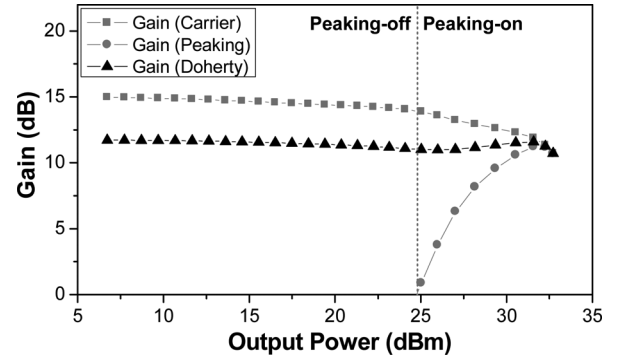


Fig. 8. Simulated gain of the power stage under Doherty operation.

As in (10), $a_1 v$ is the linear gain and the other terms represent the gain deviation of the amplifier. When the gain expansion occurs, the sign of a_3/a_1 and a_5/a_1 is positive, and the sign of the coefficient of $\cos(2w_{1,2} - w_{2,1})t$, as in (11), is the same as that of a_1 . The IM3 and the carrier signal are thus in-phase in the gain expansion amplifier such as a peaking amplifier. When gain compression occurs, as in a carrier amplifier, the IM3 and the carrier signal are in antiphase according to the same principle. When the complementary gain characteristics of the carrier and peaking amplifiers are employed, the IMD3s cancel each other out, thereby improving the linearity of the

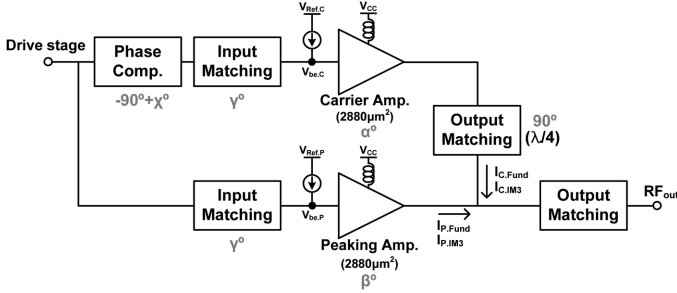


Fig. 9. Block diagram of Doherty stage.

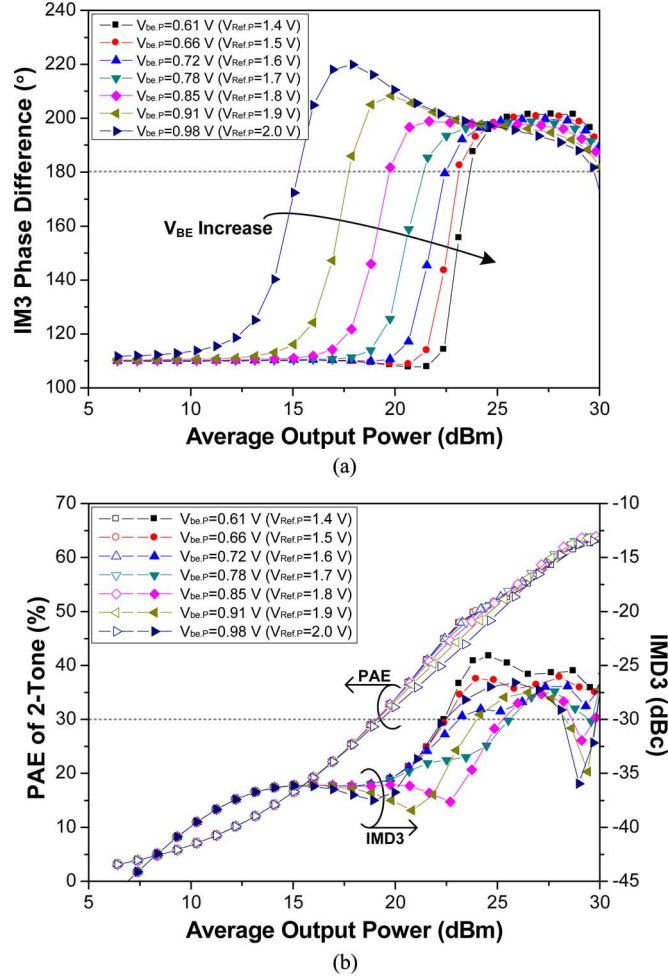


Fig. 10. Two-tone simulation results with various base biases of the peaking amplifier. (a) Phase differences of IM3 currents between the carrier and peaking amplifiers. (b) PAE and IMD3 curves versus the power sweep.

Doherty PA. These principles cancel the symmetric IMD3 only because the memory effect is not involved in this process. Fig. 9 shows a block-diagram of the proposed Doherty PA. The fundamental currents of the carrier amplifier ($I_{C,Fund}$) and the peaking amplifier ($I_{P,Fund}$) should be in-phase, but the IM3 currents of the carrier amplifier ($I_{C,IM3}$) and the peaking amplifier ($I_{P,IM3}$) should be in antiphase for achieving a good linearity. In the linear Doherty PA design, the IMD3s from the carrier and peaking amplifiers should cancel each other properly by adjusting the large-signal third-order transconductance coefficient ($gm3$), which is performed by proper base biases of the two amplifiers [12], [20], [21]. Fig. 10(a) shows the phase differences between $I_{C,IM3}$ and $I_{P,IM3}$ according to

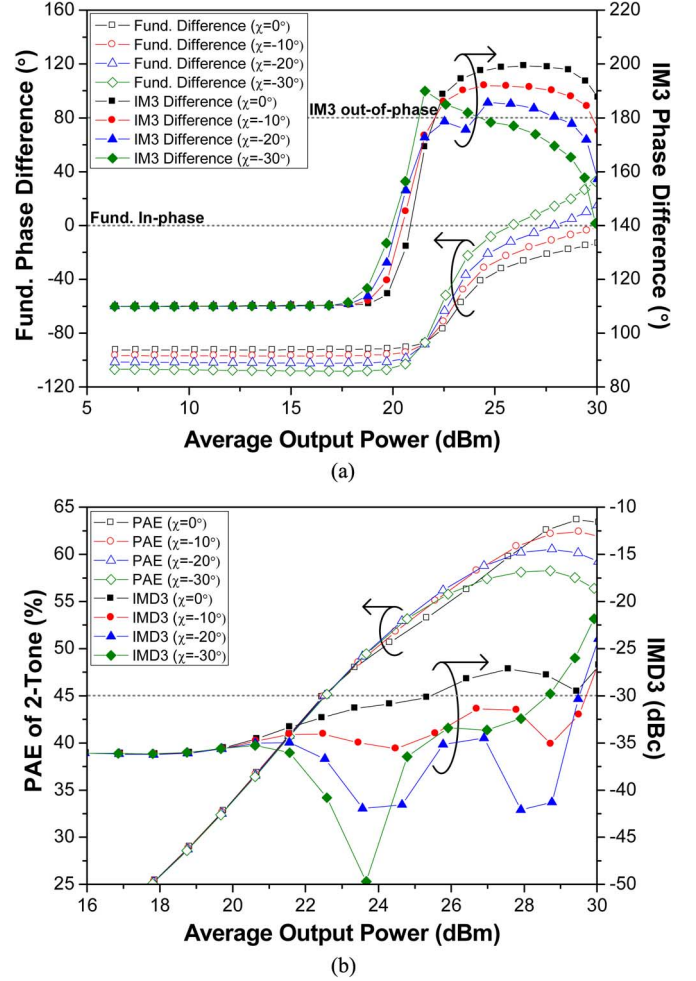


Fig. 11. Two-tone simulation results with additional phase delay at the input of the carrier amplifier. (a) Phase differences of IM3 current between the carrier and peaking amplifiers. (b) PAE and IMD3 curves versus the power sweep.

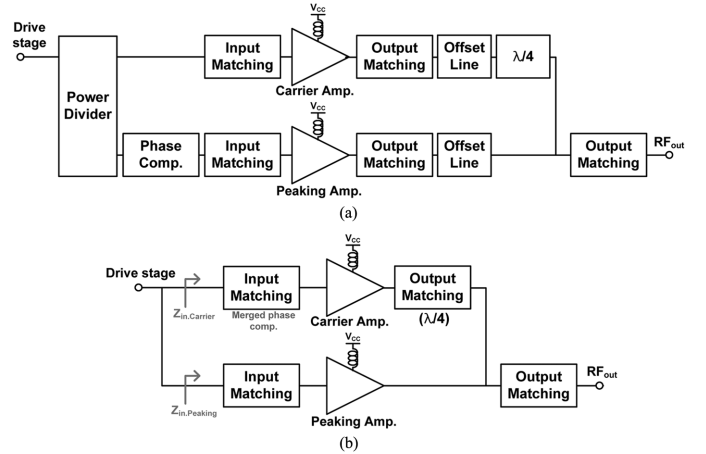


Fig. 12. Block diagram of the: (a) conventional and (b) proposed Doherty structure.

various base biases of the peaking amplifier. In this simulation, the phase compensation circuit and the output matching of the carrier amplifier ($\lambda/4$) have phases of -90° and 90° , respectively, and the input matching circuits of both amplifiers are the same. The cancellation points are varied according to the base bias points of the peaking amplifier, as shown in Fig. 10(b). The efficiency and linearity of 6-dB back-off

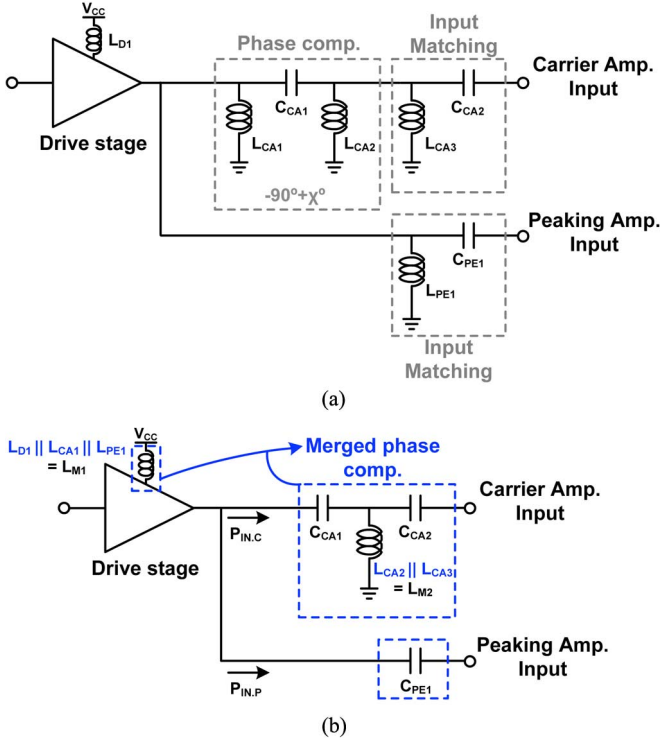


Fig. 13. (a) Schematic of the input-matching circuit for the Doherty stage with the direct-input-power-dividing technique. (b) Simplified final schematic using the merged inductor configuration.

regions are closely connected to turn-on timing of the peaking amplifier. Since the peaking amplifier cannot abruptly turn on, it should be turned on early to cancel the IM3 of the carrier amplifier for achieving a high linearity at the 6-dB back-off regions, but the early turn on means that early load modulation from $2R_{opt}$ to R_{opt} , decreasing the peak efficiency at the 6-dB power back-off (PBO). However, there is an additional phase difference of about 20° caused by the memory effect produced by the fundamental-frequency power and the feed-backed even-order harmonic or even-order intermodulation power. The two IM3 currents should be out-of-phase to optimally cancel each other. However, the fundamental currents should be in-phase for proper power combining. There is a tradeoff relationship between these currents, which should be optimized as shown in Fig. 11. In this study, the base of the peaking amplifier is biased ($V_{be,P}$) at 0.75 V and a -20° of additional phase-delay line (χ°) is inserted at the input of the carrier amplifier for both linearity and efficiency. As shown in Fig. 11, the triangle-symbol line represents the additional phase of -20° to compensate the phase delay caused by memory effects. As shown in this figure, the antiphase relationship achieves good IMD3 performance.

D. Compact Design for Handset Applications

Compact design of the Doherty PA is essential for handset applications. Fig. 12(a) shows a conventional Doherty structure, which uses a power divider such as a coupler for the input of both amplifiers. The proposed Doherty PA is implemented with a direct-input-power-dividing technique by considering the impedance variations of the carrier and peaking amplifiers, as shown in Fig. 12(b). The carrier amplifier receives more

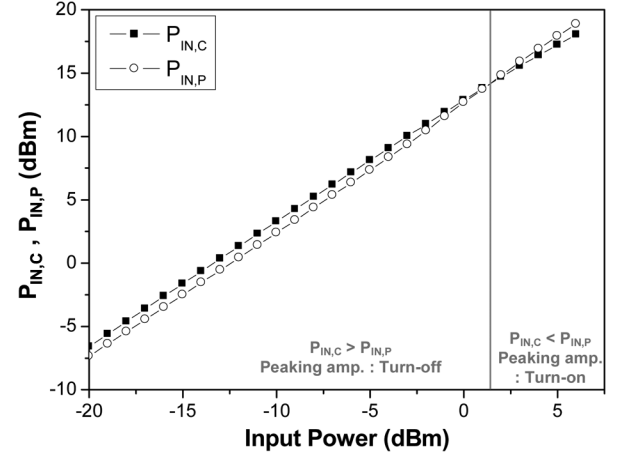


Fig. 14. Simulated results of proper uneven power driving with the simplified circuit.

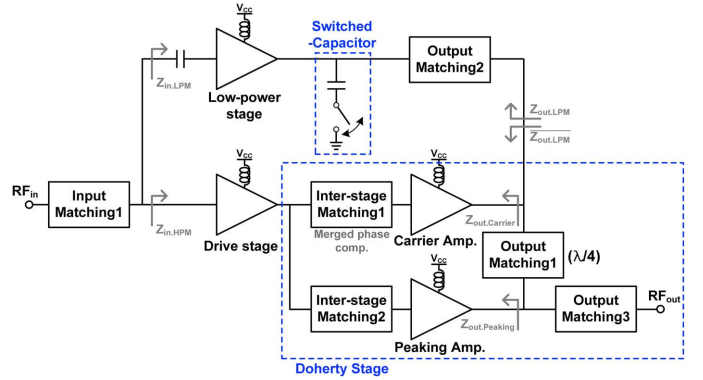


Fig. 15. Block diagram of the proposed PA.

power in the low-input power region; the peaking amplifier has more power in the high-input power region, which is the desired input drive [12]. Integrating the Doherty structure into a single chip is a difficult task. Therefore, the number as well as the values of the components should be minimized. The proposed Doherty stage shares the drive stage, which is different from those described in [11] and [12], to reduce the chip area. High-pass filter (HPF)-type input matching circuits are used for proper direct-input-power dividing [12], and a quarter-wave transformer for phase compensation is implemented in a Pi-network using two shunt inductors, as shown in Fig. 13(a). Through rearrangement of the matching components, the five inductors can be merged into two inductors, as shown in Fig. 13(b), and both the number and size of the inductors are reduced owing to the parallel structure. L_{M1} and L_{M2} are easily implemented by a bias line of the drive stage and a bondwire, respectively. With the merged-inductor structure, the Doherty stage achieves proper uneven power-input driving, as shown in Fig. 14. Both the number and values of the components are significantly reduced as compared with the conventional Doherty structure [10]–[12], [19] and only one capacitor (C_{PE1}) and one Pi-circuit are employed to drive the Doherty stage along with interstage matching.

III. ENHANCED BACK-OFF EFFICIENCY

In this section, we propose a low-power mode (LPM) for operation below the 10-dB back-off level, and Fig. 15 shows a

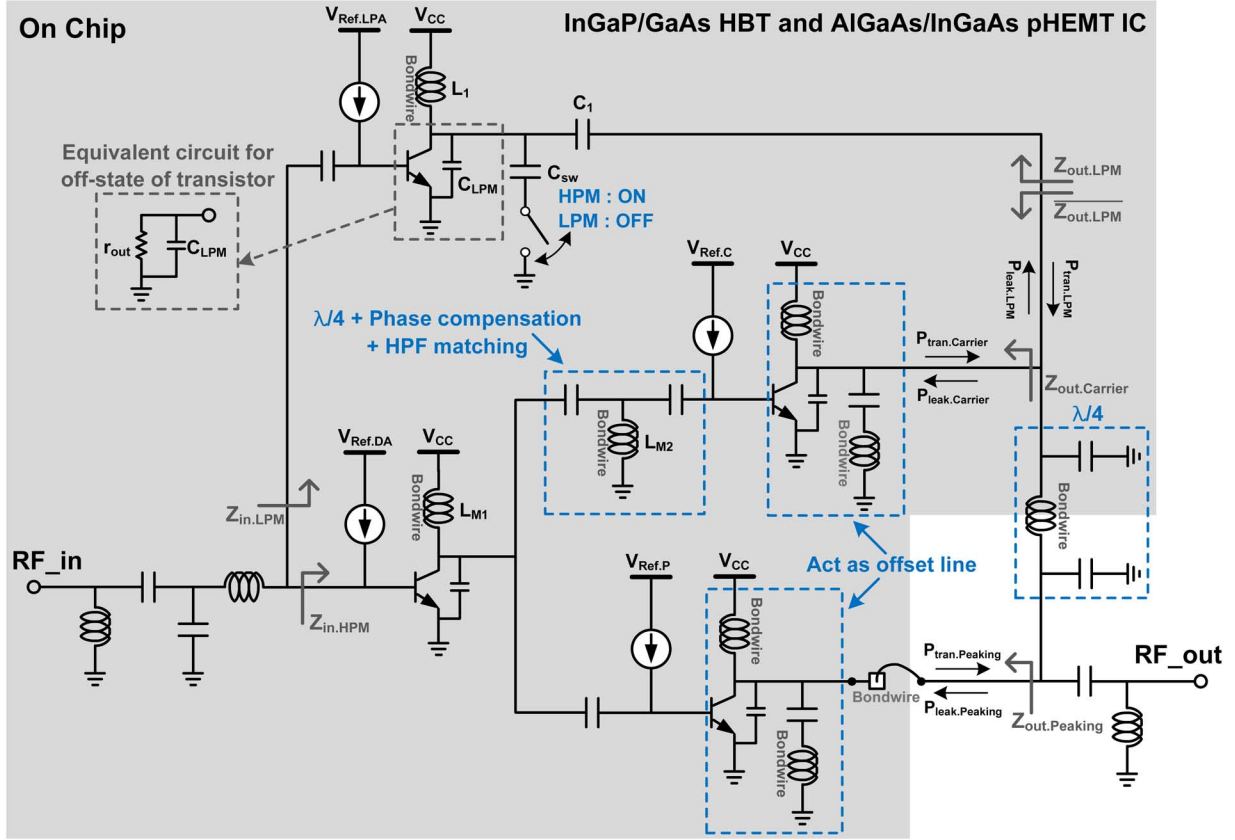


Fig. 16. Schematic of the proposed PA.

block diagram of the proposed structure. The Doherty PA architecture with a high-power mode (HPM) and an enhanced LPM is a powerful approach for mobile handset applications. In our previous work, a mode control method using a shunt switched capacitor was proposed [22], and it is adopted in this architecture. Fig. 16 shows the entire schematic of the proposed PA. This section explains the application of the control method to the proposed PA and presents an analysis based on mathematical calculations.

A. Selection of the Input Path

The input path is selected between the $Z_{in,LPM}$ and $Z_{in,HPM}$ paths by turning on/off V_{Ref} of the base bias circuits, as shown in Fig. 16. C_{in} of an HBT increases dramatically as the base voltage and size of the transistor increase, as shown in Fig. 17(a). The variations in the input capacitance in the on/off states are about 700%, which achieves an input impedance ratio of 7:1 between the off- and on-state (class-AB biased) of the transistor, as shown in Fig. 17(b). The 7:1 ratio is maintained regardless of size of the transistor, and it is used for selecting the input path for mode change. In this work, the drive stage and the low-power stage employ the same-size transistors of $720 \mu m^2$. The input matching of the LPM requires a series capacitor to block the dc base bias between the drive stage and the low-power stage. The on/off states barely affect the matching and are used to select the input path, as shown in Fig. 17(c), because of the high-impedance ratio of 7:1. No separate switch is required for signal path selection with a loss lower than 0.5 dB [22] because of the input capacitor

characteristics. On the other hand, the input impedance ratio of 7:1 in the Doherty mode can be manipulated with the HPF-type input matching and the phase compensation circuit. Therefore, the input impedances of the carrier and peaking amplifiers are adjusted for proper uneven input-power driving as the input power increases, as described in the previous sections [see Fig. 17(d)].

B. Dual-Power-Mode Operation

For proper HPM operation, the outputs of the carrier and peaking amplifiers should not leak into the $Z_{out,LPM}$ path. The shunt switched capacitor at the LPM path increases $Z_{out,LPM}$, thereby reducing the leakage loss. In the HPM, the off state of the lower power stage is expressed by C_{LPM} in parallel with r_{out} , as shown in Fig. 16. Without the switch loss, $Z_{out,LPM}$ is expressed as (where r_{out} is assumed to be infinity)

$$Z_{out,LPM} = \left(\frac{1}{j\omega_0(C_{LPM} + C_{sw})} \parallel j\omega_0 L_1 \right) + \frac{1}{j\omega_0 C_1} \quad (12)$$

where

$$L_1 = \frac{Z_{opt,LPM}}{\omega_0 \cdot \sqrt{m} - 1} \quad (13)$$

$$C_1 = \frac{1}{\omega_0 \cdot \sqrt{m} - 1 \cdot Z_{out,LPM}} \quad (14)$$

$$m = \frac{Z_{opt,LPM}}{Z_{out,LPM}}. \quad (15)$$

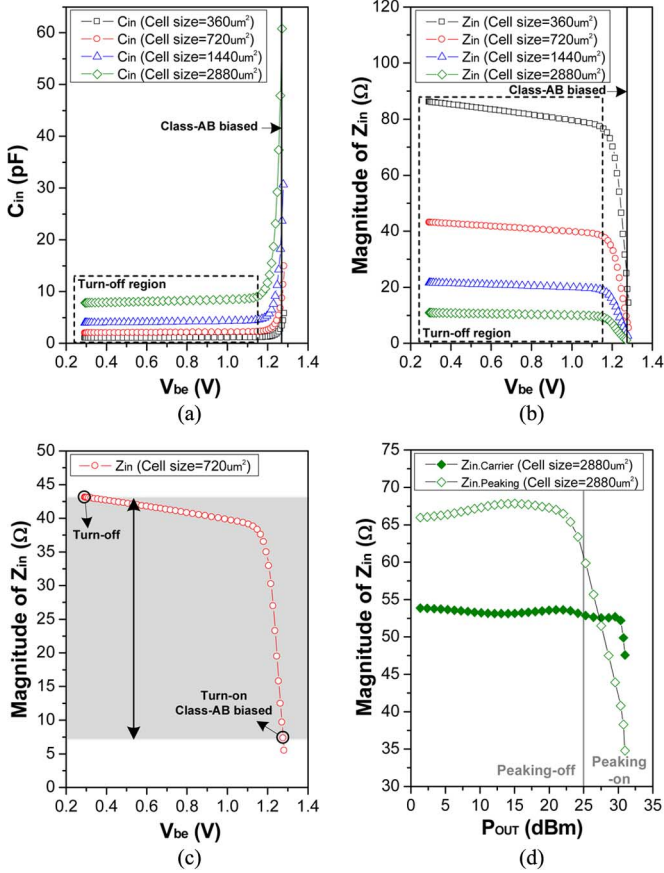


Fig. 17. Input capacitance variation with base bias voltage. (a) Value of the input capacitance and (b) input impedance versus base bias voltage with various transistor sizes. (c) Input impedance ratio between off and on states of the transistor. (d) Manipulated input impedance for the Doherty stage.

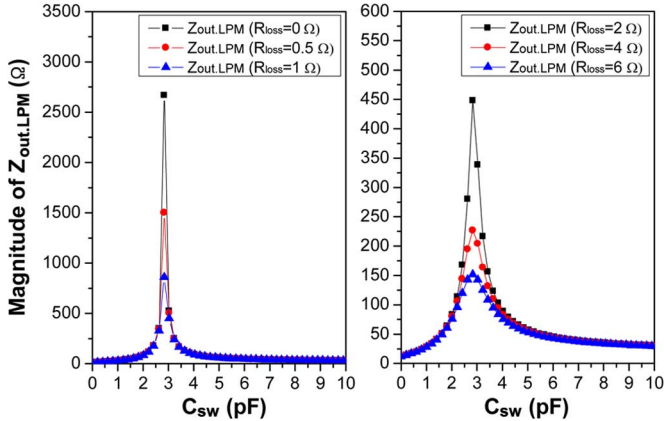


Fig. 18. Calculated magnitude of $Z_{out.LPM}$ versus the value of switched capacitor (C_{sw}) in the HPM considering the switch losses (R_{loss}).

Therefore, the magnitude of $Z_{out.LPM}$ is given,

$$|Z_{out.LPM}| = \frac{w_0 \cdot C \cdot Z_{opt.LPM} \cdot \overline{Z_{out.LPM}} \sqrt{m-1}}{w_0 \cdot C \cdot Z_{opt.LPM} - \sqrt{m-1}} + \frac{Z_{opt.LPM} - \overline{Z_{out.LPM}}(m-1)}{w_0 \cdot C \cdot Z_{opt.LPM} - \sqrt{m-1}},$$

$$C = C_{LPM} + C_{sw} \quad (16)$$

where w_0 is the center frequency and $Z_{opt.LPM}$ is the optimum impedance for the low-power stage. From the VBIC model, C_{LPM} is found to be 0.425 pF. With the proper $Z_{opt.LPM}$ and $\overline{Z_{out.LPM}}$ in the LPM, the magnitude of $Z_{out.LPM}$ in the HPM

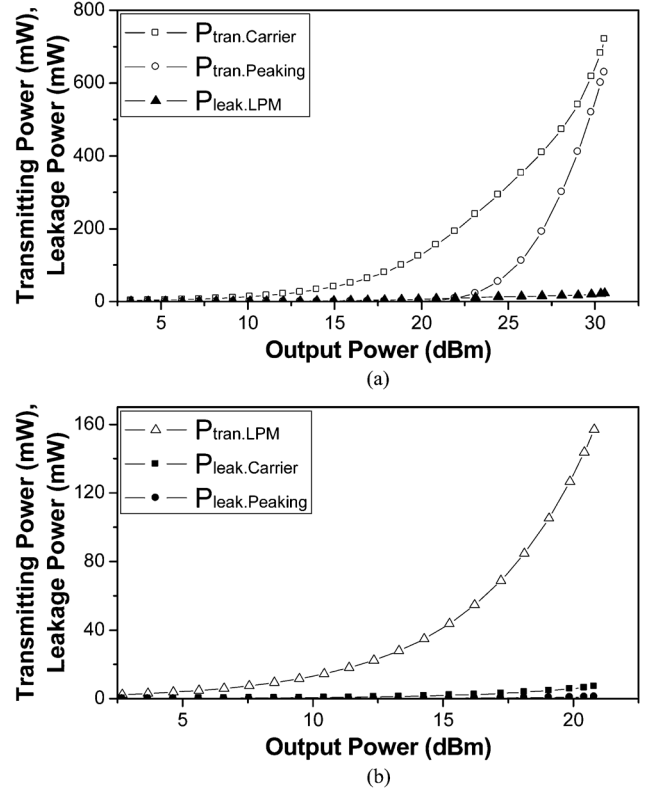


Fig. 19. CW simulated results of transmitting power (P_{tran}) and leakage power (P_{leak}) in: (a) HPM and (b) LPM operations.

TABLE I
BIAS CONDITIONS FOR DUAL POWER MODE

	$V_{Ref.LPA}$	$V_{Ref.DA}$	$V_{Ref.C}$	$V_{Ref.P}$	C_{SW}
HPM	off	on	on	on	on
LPM	on	off	off	off	off

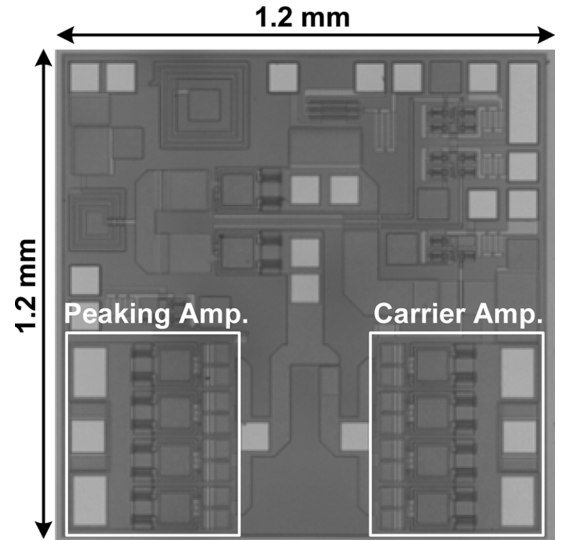


Fig. 20. Photograph of a fabricated chip of the proposed PA.

is calculated as a function of C_{sw} , as shown in Fig. 18. In this study, the switch resistance $R_{loss} < 4 \Omega$, and the magnitude of $Z_{out.LPM}$ is greater than 200 Ω when the optimum $C_{sw} = 3$ pF is used. This impedance is about 20 times greater than that of the output path, thereby preventing power leakage, as shown in

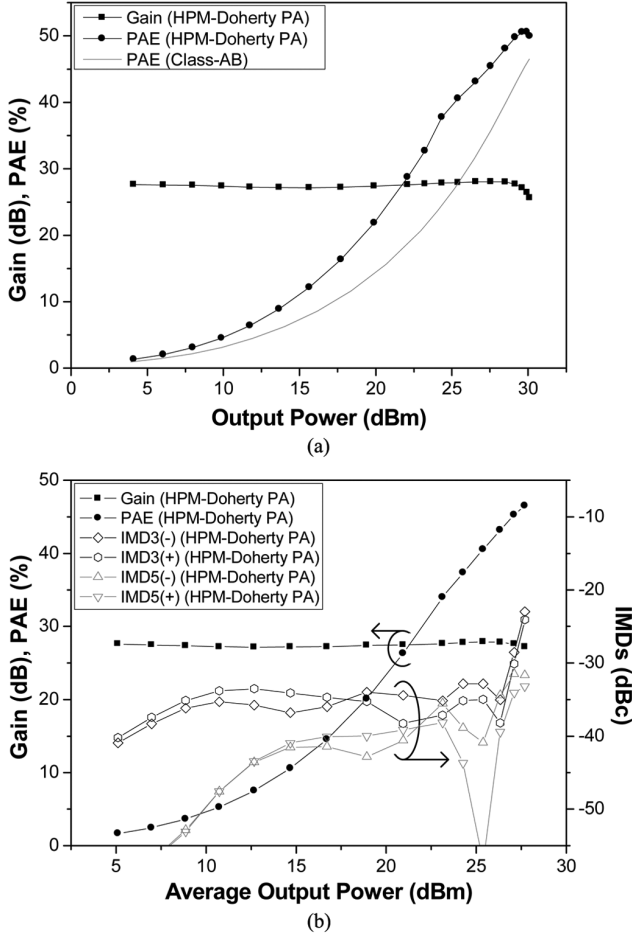


Fig. 21. Measured HPM performance of the proposed dual-power-mode Doherty PA with 3.5-V V_{cc} . (a) Gain, PAE, and IMD performances versus the power sweep for the: (a) CW signal and (b) two-tone signal.

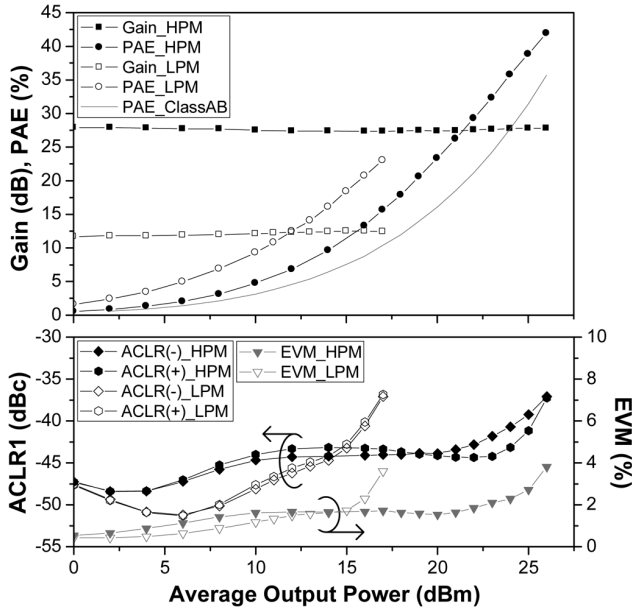


Fig. 22. Measured performance of the proposed PA with 16-QAM 7.5-dB PAPR LTE signal. The Gain, PAE, EVM, and $ACLR1_{UTRA}$ performances versus the power sweep.

Fig. 19(a). The transmitting powers of the carrier and peaking amplifiers show the proper behavior of Doherty operation. In LPM operation, the output of the low-power stage should not

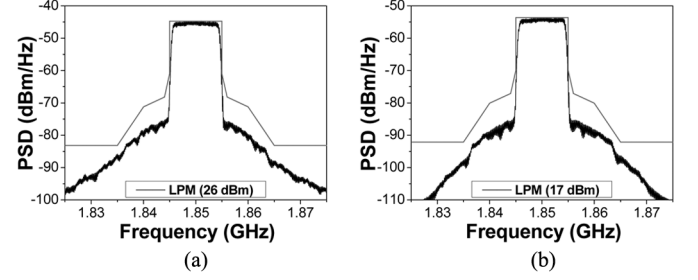


Fig. 23. Measured LTE spectrum of the PA at an: (a) average output power of 26 dBm in the HPM and (b) average output power of 17 dBm in the LPM.

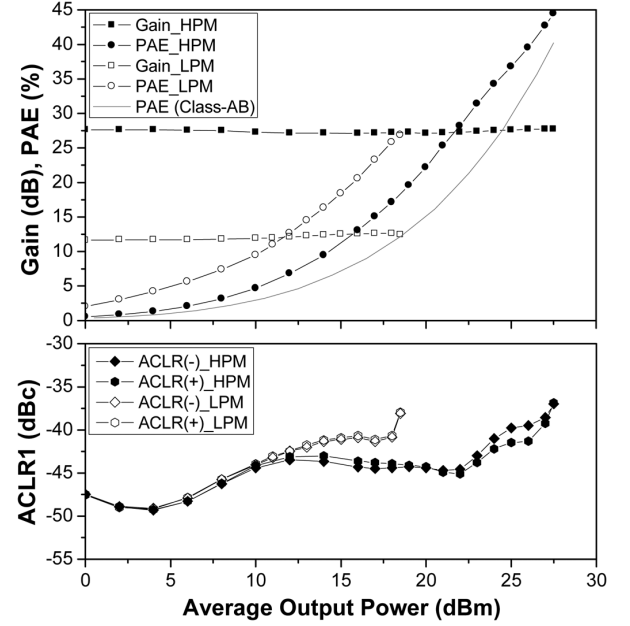


Fig. 24. Measured performance of the proposed PA with 3.3-dB PAPR WCDMA signal. The Gain, PAE, and $ACLR1_{UTRA}$ performances versus the power sweep.

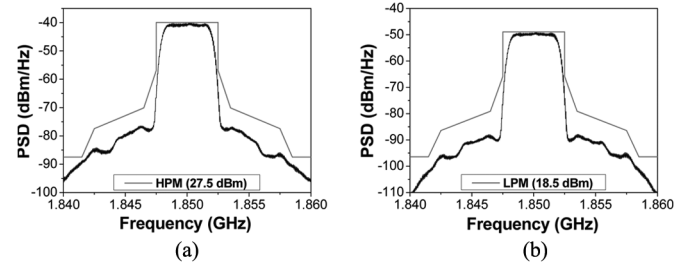


Fig. 25. Measured WCDMA spectrum of the PA at an: (a) average output power of 27.5 dBm in the HPM and (b) average output power of 18.5 dBm in the LPM.

leak into the $Z_{out,Carrier}$ as well as $Z_{out,Peaking}$ path. The output capacitances of the transistors in the Doherty cells and the series LC circuit are resonated out by the inductance of the bias line. Therefore, the imaginary parts of $Z_{out,Carrier}$ and $Z_{out,Peaking}$ is eliminated, thereby achieving high impedance and preventing power leakage, as shown in Fig. 19(b). The bias conditions for each power mode are summarized in Table I.

IV. IMPLEMENTATION AND MEASUREMENT

The proposed Doherty PA with enhanced back-off efficiency mode is fabricated by co-integration of an InGaP/GaAs HBT

TABLE II
PERFORMANCE COMPARISON OF RECENTLY REPORTED LINEAR HANDSET PAs

Ref.	Technology	Characteristic	Signal/PAPR	Frequency	Pout (dBm)	PAE (%)	ACLR1 _{UTRA} * (dBc)
APMC 2006 [23]	HBT	Doherty PA	WCDMA/–	1.95 GHz	26.7	41	–38
					6-dB PBO	26.5 [†]	–38 [†]
					17	20 [†]	–40 [†]
TMTT 2012 [24]	HBT	Multi-Band, Load-switching PA	WCDMA/–	1.85 – 1.91 GHz	28.5	40.7	–39.1
					6-dB PBO	21 [†]	–44 [†]
					16	18 [†]	–47 [†]
This work	HBT/pHEMT	Doherty combined Load-switching PA	WCDMA/3.3 dB	1.85 GHz	27.5	44.5	–37
					6-dB PBO	27.5	–44.7
					18.5	27	–38
BCTM 2011 [5]	HBT/pHEMT	Load-switching PA	LTE(QPSK)/–	1.92 – 1.98 GHz	27.5	38	–39
					6-dB PBO	15 [†]	–48 [†]
					16	24	–39
IMS 2012 [25]	HBT & CMOS	ET PA	LTE(16QAM)/6.44 dB	1.74 GHz	27	40.2	–39.6
					6-dB PBO	23.2	–38.5
					18	16.1	–39
This work	HBT/pHEMT	Doherty combined Load-switching PA	LTE(16QAM)/7.5 dB	1.85 GHz	26	42	–37
					6-dB PBO	23.4	–44
					17	23	–37

*st WCDMA and LTE specifications: ACLR1_{UTRA} < –33 dBc

[†] graphically estimated

and an AlGaAs/InGaAs pseudomorphic high electron-mobility transistor (pHEMT) switch. The switched capacitor for LPM/HPM control is built using the pHEMT device. The two capacitors at the output are external components. The inductors are replaced by bondwires and spiral inductors on the chip. The chip of the PA is mounted on an FR-4 printed circuit board. A photograph of chip with dimensions of 1.2 mm × 1.2 mm is shown in Fig. 20. Fig. 21 shows the measured HPM performance of the PA with a supply voltage of 3.5 V at the 1.85-GHz frequency band. Doherty operation achieves not only flat gain, but also good IMDs by using the techniques described in Section II.

For multi-mode demonstrations, the proposed linear PA is tested at 1.85 GHz using LTE as well as WCDMA signals. Fig. 22 shows the measured performance for a 10-MHz BW, 16-quadrature amplitude modulation (QAM), and 7.5-dB PAPR LTE signal. The PA in the HPM with Doherty operation delivers a gain of 27.8 dB, a PAE of 42%, an error vector magnitude (EVM) of 3.8%, and an access adjacent channel leakage ratio (ACLR1_{UTRA}) of –37 dBc at an average output power of 26 dBm. For a high PAPR signal such as LTE, these results demonstrate convincingly that Doherty operation is effective in achieving high efficiency in the peak output power as well as in the back-off power region. In the LPM, the PA delivers a gain of 12 dB, a PAE of 23%, an EVM of 3.6%, and an ACLR1_{UTRA} of –37 dBc at an average output power of 17 dBm. Fig. 23 shows the measured spectra, satisfying the standard spectrum mask at an average output power of 26 dBm for the HPM and 17 dBm for the LPM. For a 3.84-MHz BW and 3.3-dB PAPR WCDMA

signal, as shown in Fig. 24, the PA achieves a gain of 27.7 dB, a PAE of 44.5%, and an ACLR1_{UTRA} of –37 dBc at an average output power of 27.5 dBm in the HPM and a gain of 12 dB, a PAE of 27%, and an ACLR1_{UTRA} of –38 dBc at an average output power of 18.5 dBm in the LPM. Fig. 25 shows the measured spectra, satisfying the standard spectrum mask at an average output power of 27.5 dBm for the HPM and 18.5 dBm for the LPM. Table II summarizes the comparison of the measured results with the previously reported results of the linear handset PAs. The proposed PA delivers higher efficiency at all output power levels especially at the peak output power and 6-dB PBO regions compared with recent works. When the Doherty mode is operated for the LTE signal, the proposed standalone PA delivers a similar efficiency curve for an envelope tracking (ET) PA across the overall output power range.

V. CONCLUSIONS

A linear Doherty PA with enhanced back-off efficiency mode has been developed. A detailed analysis of the Doherty operation for gain modulation and IMD3 cancellation are investigated to achieve a highly linear performance. A compact design method has been proposed to overcome the size constraint for handset applications. The enhanced back-off efficiency mode with proper Doherty operation was adopted by using a shunt switched capacitor, resulting in high efficiency across the overall output power range for LTE as well as WCDMA signals. For a 16-QAM LTE signal with 7.5-dB PAPR and 10-MHz BW, the dual-power-mode Doherty PA achieved a gain of 27.8 dB, a PAE of 42%, an EVM of 3.8%,

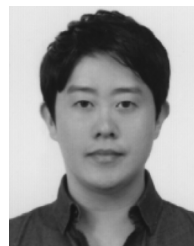
and an ACLR_{1UTRA} of -37 dBc at an average output power of 26 dBm in the HPM with the Doherty operation and a gain of 12 dB, a PAE of 23%, an EVM of 3.6%, and an ACLR_{1UTRA} of -37 dBc at an average output power of 17 dBm in the LPM. For a 3.84-MHz BW and 3.3-dB PAPR WCDMA signal, the PA achieved a gain of 27.7 dB, a PAE of 44.5%, and an ACLR_{1UTRA} of -37 dBc at an average output power of 27.5 dBm in the HPM and a gain of 12 dB, a PAE of 27%, and an ACLR_{1UTRA} of -38 dBc at an average output power of 18.5 dBm in the LPM. These results indicate that the Doherty PA with the LPM configuration is a promising candidate for extending the battery life of handheld devices in 3G as well as 4G wireless communication devices.

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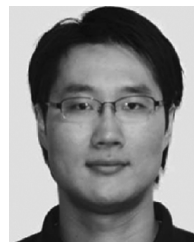
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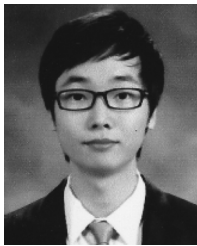
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